**MSBA 265: Business Analytics Topics**

**Foundational Module 1 - Practical Homework Assignment**

Interactive EDA, Data Quality Verification,

Data Dictionaries, & Outlier Pipelines

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## Part 1 - Workspace Setup, Git Ingestion, & Procurement

The project was organized as a Git-tracked VS Code workspace using a Python virtual environment. The repository contains the required data, notebook, source-code, and report directories, with a .gitignore configured to exclude the virtual environment, Python cache files, macOS metadata, and Jupyter checkpoint files.

| Required Artifact | Project Location |
|---|---|
| Ingestion script | data/download_data.py |
| Raw dataset | data/raw_business_data.csv |
| EDA notebook | notebooks/01_eda_and_data_dictionary.ipynb |
| Outlier pipeline | src/clean_outliers.py |
| Data dictionary | reports/data_dictionary.csv |
| Visual artifacts | reports/figures/ |

**Reproducibility note**

The repository should be runnable from a clean environment by creating/activating a virtual environment, installing requirements.txt, running the ingestion script, executing the notebook, and then running the production outlier script.

## Part 2 - Raw Boundary Audits & Business Data Dictionary

### 2.1 Structural Data Types & Summary Statistics

| Feature | Min | Q1 | Median / Q2 | Max |
|---|---|---|---|---|
| DrivAge | 18 | 34 | 44 | 100 |
| Exposure | 0.002732 | 0.18 | 0.49 | 2.01 |
| BonusMalus | 50 | 50 | 50 | 230 |
| Density | 1 | 92 | 393 | 27,000 |

The notebook executes df.info() and df.describe().T to verify structure and numerical summaries. The dataset contains 678,013 records and 12 variables, and all 12 variables show 678,013 non-null values; therefore, no missing-value remediation is required at this stage.

**2.2 Raw Boundary Audit**

The raw boundary audit confirms that the numerical variables are within plausible ranges. DrivAge ranges from 18 to 100 years, so there are no negative-age records. Exposure ranges from approximately 0.0027 to 2.01, with no negative exposure values. BonusMalus ranges from 50 to 230, with no negative or sentinel-coded values. Overall, the audited variables do not show obvious negative-value corruption or sentinel null codes.

**2.3 Business Data Dictionary Artifact**

The notebook builds a business data dictionary for all 12 fields and exports the required artifact to reports/data_dictionary.csv, including business labels, data types, units, null counts, unique-value counts, and business definitions.

**2.4 Business Rationale: Exposure Offset and Density/Area Collinearity**

Exposure represents the amount of time an insurance policy is active. For count-based claim-frequency modeling, log(Exposure) should be incorporated as an offset rather than treating Exposure as a standard predictor. A policy observed for a longer period naturally has more opportunity to generate claims; therefore, claim frequency should be modeled relative to exposure time. In addition, continuous Density and categorical Area should not both be included in the same linear regression model because they capture overlapping geographic risk information. Including both may introduce redundancy and multicollinearity, which can inflate coefficient variance and reduce model stability and interpretability. Therefore, Density should be retained as the continuous geographic predictor while Area can be excluded.

# Part 3 - Automated Rule Override & Correlation Audit

## 3.1 Pearson Correlation Heatmap

reports/figures/correlation_heatmap.png


***Figure 1. Pearson correlation heatmap generated from the numerical features.***

- **Distribution/Relationship Shape:** The numerical feature relationships are mostly weak to modest; there is no pair with a near-perfect linear relationship.
- **Key Metrics/Whiskers:** The strongest visible off-diagonal relationship is DrivAge vs. BonusMalus at approximately r = -0.48. The automated audit reported no severe pair with |r| >= 0.85.

- **Recommended Analytics Action:** No numerical feature needs to be removed solely because of severe Pearson multicollinearity. Domain redundancy should still be handled separately, especially for geographic variables such as Density and Area.

### 3.2 ClaimNb Override

Automated skewness diagnostics flag ClaimNb as extremely right-skewed (skewness approximately 5.60) and would normally recommend a logarithmic transformation. That recommendation is intentionally overridden. ClaimNb is a discrete, non-negative count variable with a very large mass at zero; applying a natural logarithm to exact zeros is mathematically undefined (log(0) tends to negative infinity). The business meaning also matters: ClaimNb represents event counts, not continuous volume. It should remain in count units and be modeled as a claim-frequency target using a Poisson rate framework, with log(Exposure) incorporated as an offset.

## Part 4 - Distribution Audits & Written Plot Statements

reports/figures/feature_distributions.png

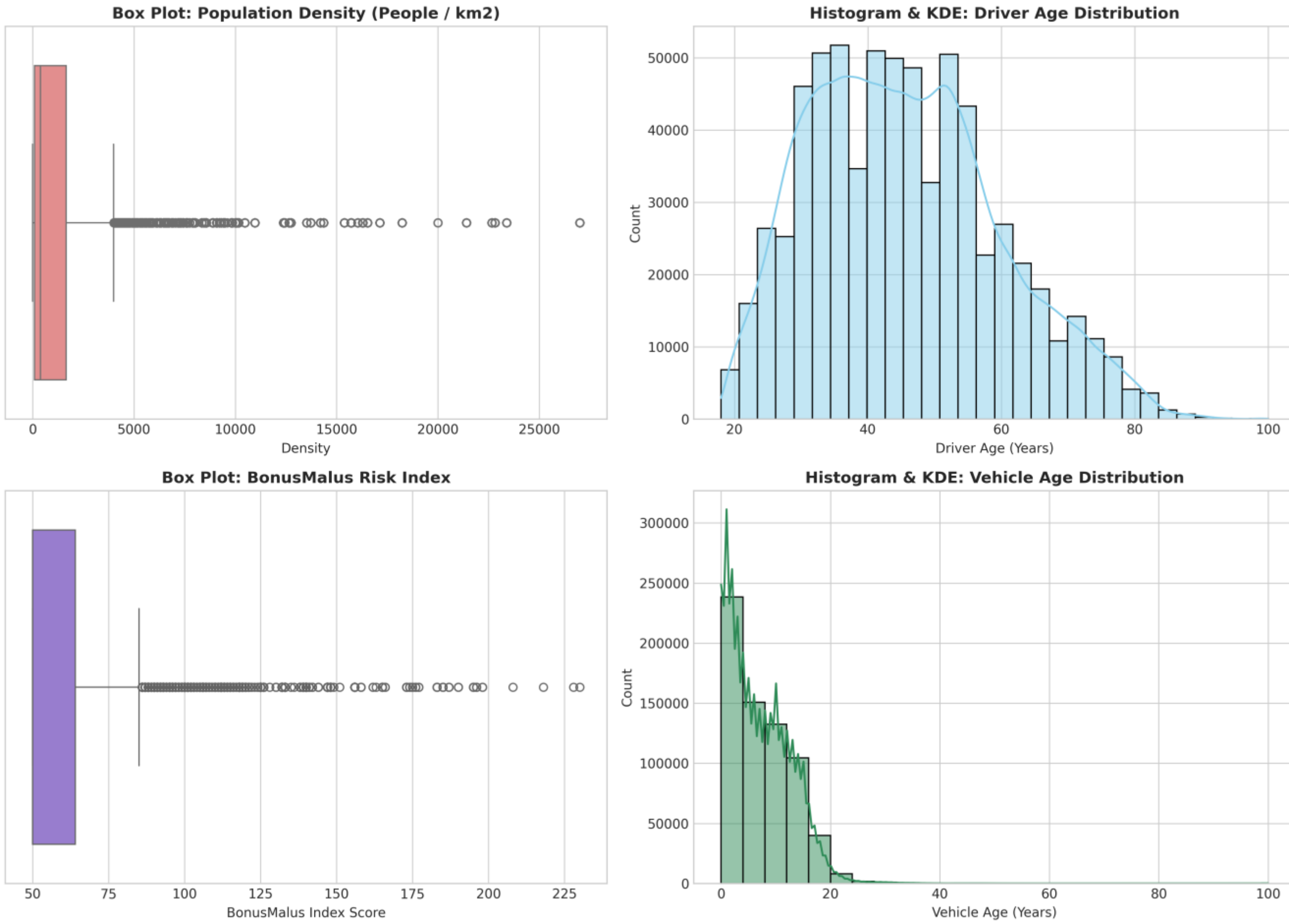


*Figure 2. Distribution audit: Density, DrivAge, BonusMalus, and VehAge.*

### 4.1 Density box plot

- **Distribution/Relationship Shape:** Density is extremely right-skewed, with a long upper tail representing highly urban municipalities.
- **Key Metrics/Whiskers:** Q1 = 92, median = 393, and Q3 = 1,658 people/km². The Tukey upper fence is approximately 4,007 people/km²; values above this boundary are statistical outliers even when they are valid urban records.
- **Recommended Analytics Action:** Do not automatically discard high-density policyholders because they may represent legitimate metropolitan observations. Retain these records and apply log(1 + Density) before linear regression to reduce the influence of the strong right skew.

**4.2 Driver age histogram & KDE**

- **Distribution/Relationship Shape:** DrivAge is comparatively balanced and only mildly skewed (skewness approximately 0.44), with the mass centered in the middle adult-age range.
- **Key Metrics/Whiskers:** Q1 = 34, median = 44, Q3 = 55, and the observed range is 18 to 100 years.
- **Recommended Analytics Action:** Retain the full valid age range. Standard scaling is sufficient for models that benefit from normalized continuous inputs; no routine outlier trimming is required.

**4.3 BonusMalus box plot**

- **Distribution/Relationship Shape:** BonusMalus is right-skewed and concentrated at the lower end, with a large mass at the baseline score of 50 and a smaller high-risk tail.
- **Key Metrics/Whiskers:** Q1 = 50, median = 50, Q3 = 64, and the observed maximum is 230. Scores above the upper whisker represent high-risk penalty observations rather than automatically implying data corruption.
- **Recommended Analytics Action:** Retain and scale BonusMalus as a continuous risk variable and create a binary feature Is_Surcharge = BonusMalus > 100 to identify high-risk penalty drivers.

**4.4 Vehicle age histogram & KDE**

- **Distribution/Relationship Shape:** VehAge is moderately right-skewed (skewness approximately 1.15), with many newer vehicles and a diminishing long tail toward older vehicles.
- **Key Metrics/Whiskers:** Q1=2 years, median=6 years, Q3= 11 years, with an observed maximum of 100 years.
- **Recommended Analytics Action:** Retain VehAge and bin it into practical age brackets (0–5 years, 6–10 years, and 11+ years), or apply min-max scaling for downstream modeling.

## Part 5 - Production Outlier Pipeline & Final Deliverable

### 5.1 Production Outlier Pipeline: Density Filtering Results

reports/figures/outlier_filtering_comparison.png

| Method | Rule / Valid Range | Records Excluded | Interpretation |
|---|---|---|---|
| Tukey IQR | -2,257 to 4,007 | 77,566 (11.44%) | Aggressive on the long right tail |
| Z-score | \|Z\| <= 3.0 | 15,209 | Less restrictive because skewness inflates mean and SD |

**Original records:** 678,013 | **Final IQR-cleaned records:** 600,447

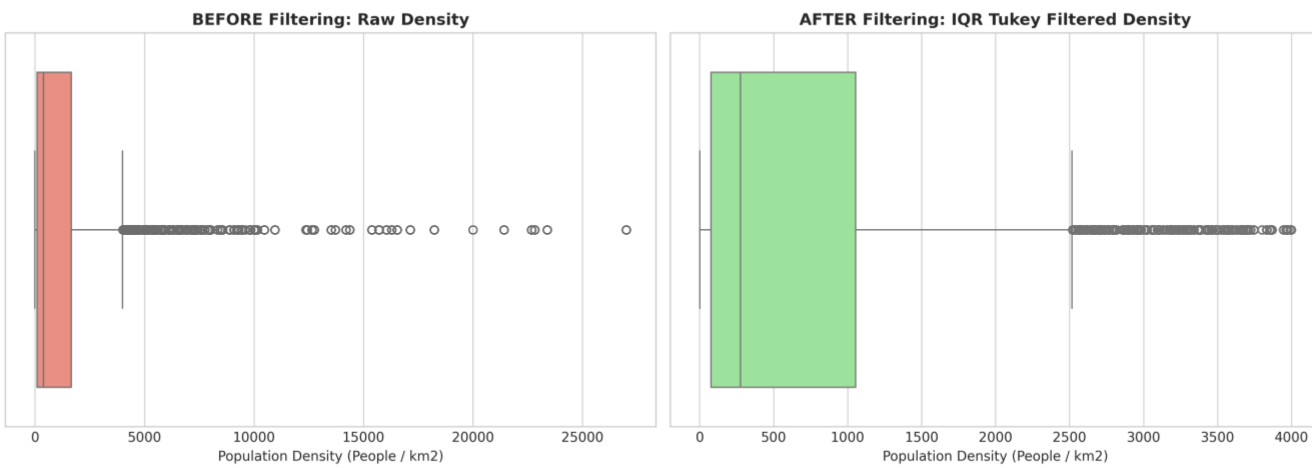


***Figure 3. Before/after Density box plots from the Tukey IQR filtering workflow.***

### 5.2 Final Report Interpretation and Figures

The large difference between Tukey-IQR and Z-score filtering occurs because Density is highly right-skewed. Extreme metropolitan values pull the mean upward and inflate the standard deviation, which makes a Z-score cutoff much less restrictive. Tukey's quartile-based rule is more robust to skewness, but it can remove a substantial share of valid urban policyholders. Therefore, the filtering choice must follow the business objective rather than a purely mechanical outlier rule.

- **Distribution/Relationship Shape:** The raw Density distribution has a compressed central box and a long right tail; after Tukey filtering, the visual range is much more compact.
- **Key Metrics/Whiskers:** The IQR rule excludes 77,566 records (11.44%), compared with 15,209 under the Z-score rule. This is a materially different population definition.
- **Recommended Analytics Action:** For statewide risk pricing, preserve valid urban records and transform Density instead of dropping them. Use Tukey trimming only when the modeling population is intentionally restricted or when a sensitive linear model requires it.